Accepted Manuscript

A boron nitride-polyvinylidene fluoride-co-hexafluoropropylene composite gel polymer electrolyte for lithium metal batteries

Xiaofei Bian, Junfei Liang, Xiaofu Tang, Runxia Li, Lei Kang, Anyu Su, Xin Su, Yingjin Wei

PII: S0925-8388(19)32412-0

DOI: https://doi.org/10.1016/j.jallcom.2019.06.321

Reference: JALCOM 51220

To appear in: *Journal of Alloys and Compounds*

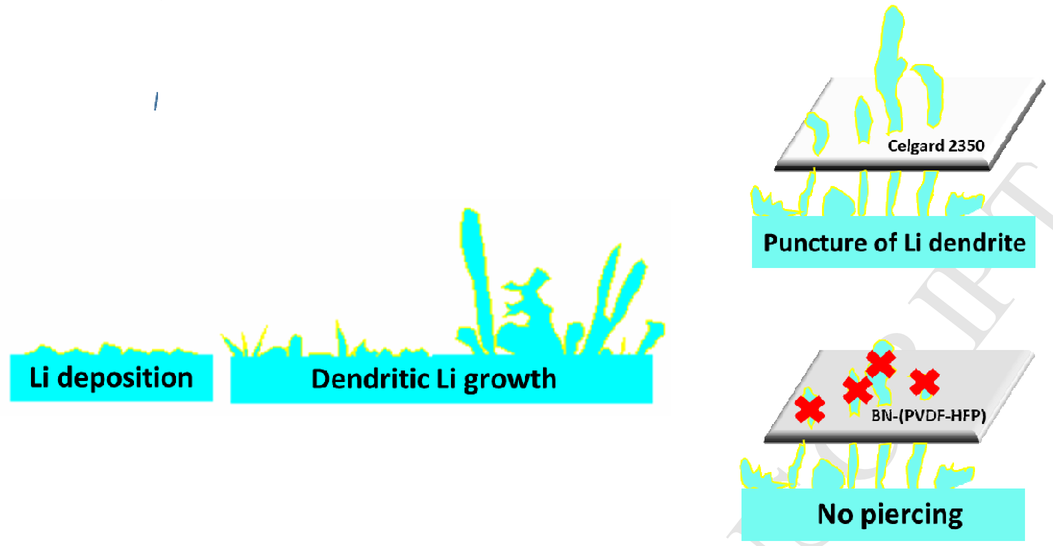
Received Date: 10 April 2019

Revised Date: 24 June 2019

Accepted Date: 26 June 2019

Please cite this article as: X. Bian, J. Liang, X. Tang, R. Li, L. Kang, A. Su, X. Su, Y. Wei, A boron nitride-polyvinylidene fluoride-co-hexafluoropropylene composite gel polymer electrolyte for lithium metal batteries, *Journal of Alloys and Compounds* (2019), doi: https://doi.org/10.1016/j.jallcom.2019.06.321.

A modified GPE with a multifunctional BN additive can revitalize LMBs to readily obtain a comparative ionic conductivity and remarkably improve mechanical modulus, which is beneficial for suppressing Li dendrite formation.



A boron nitride-polyvinylidene fluoride-co-hexafluoropropylene composite gel polymer electrolyte for lithium metal batteries

Xiaofei Bian[a], Junfei Liang[a], Xiaofu Tang[a], Runxia Li[a,*], Lei Kang[b,*], Anyu Su[c], Xin Su[c], Yingjin Wei[c,*]

[a] School of Materials Science and Engineering, Dongguan University of Technology, Dongguan 523808, China.

[b] College of Medical Information Engineering, Guangdong Pharmaceutical University, Guangzhou 510006, China.

[c] Key Laboratory of Physics and Technology for Advanced Batteries (Ministry of Education), College of Physics, Jilin University, Changchun 130012, China.

Corresponding authors:

5327368@qq.com; kanglei8801@163.com; yjwei@jlu.edu.cn

ABSTRACT

A modified gel polymer electrolyte (GPE) with boron nitride (BN) additive was designed to boost the electrochemical performance of rechargeable lithium metal battery, which consists of Li-rich layered cathode and possesses a high energy density. BN, electron insulator and ion conductor, has excellent thermodynamic and electrochemical stability. BN particles were well dispersed in polyvinylidene fluoride-co-hexafluoropropylene (PVDF-HFP) polymer matrices and thus substantially enhanced the electrochemical and physical properties of the GPEs. With only 0.5 wt% BN additive, it remarkably improved the mechanical modulus and ionic conductivity (to 4.1×10^{-4} S·cm^{-2}) of GPEs, which was beneficial for suppressing lithium dendrite formation and fast ion transportation, respectively, thereby enabling high-performance lithium metal batteries with ultra-long cycle life and high safety at ambient temperature (25 ^{o}C) and high temperature (55 ^{o}C).

Keywords

Rechargeable lithium metal battery; Li-rich layered oxide; Gel polymer electrolyte; BN additive.

1. Introduction

Consumer electronics, electrified transportation, and large-scale energy storage systems increasingly require environment-friendly energy conversion technologies. In this regard, new rechargeable batteries with high safety, long lifespan, and high energy density are urgently needed to satisfy these increasing demands. Because of its advantages such as the highest theoretical specific capacity (3860 $mA \cdot h \cdot g^{-1}$), low density (0.59 $g \cdot cm^{-3}$), and the lowest redox potential (-3.04 V vs. standard hydrogen potential), lithium metal is a promising alternative as the negative electrode for rechargeable metal batteries[1-5]. However, several remaining issues should be overcome before practical application of lithium metal batteries, including (1) the formation of destructive lithium dendrites, (2) thermodynamic instability of lithium metal, resulting in the formation of thick solid-electrolyte-interphase (SEI) film and consumption of lithium and electrolyte, and (3) volume and morphology changes during cycling. Among these issues, uncontrollable lithium dendrite growth during repeated lithium deposition/stripping causes severe safety problems due to the possible piercing of the separator such as battery short circuit or even explosion [6-9].

In recent years, much effort has been made to develop lithium metal-based rechargeable batteries (i.e. lithium metal batteries, LMBs), including (1) construction of an artificial SEI layer on lithium metal, (2) electrolyte additives with electrostatic shielding effects, and (3) applying all-solid electrolytes with high hardness and high Li^+ transference number [10-18]. Among these, all-solid electrolyte-based LMBs can effectively inhibit lithium dendrite punctures and thus, improve battery safety. However, it is still challenging to introduce these batteries into the market due to their disadvantages such as low ionic conductivity, slow ion transport and large interfacial resistance between the electrode and all-solid electrolyte. Besides, there are some safety concerns for liquid-electrolyte based LIBs, which may be attributed to the low mechanical strength and high risk of electrolyte spills. Advantageously, gel polymer electrolytes (GPEs) have been widely used as quasi-solid-state electrolytes to overcome the intrinsic shortcomings of both liquid-electrolyte and all-solid-state electrolyte-based LIBs. The polymers are acting as a supporting skeleton and quasi-solid-state electrolyte shows desirable properties of favorable film-forming, high film strength, and wide electrochemical stability window. Several polymer skeleton materials have been used in GPEs, such as polyacrylonitrile (PAN), polyoxyethylene (PEO), and polyvinylidene fluoride (PVDF). In such GPEs, lithium salts, such as $LiClO_4$, $LiBF_4$, and $LiPF_6$, have been used as electrolyte salts [19-25]. However, the bad mechanical strength and low ionic conductivity of GPEs obstacle their practical application in LMBs.

In this study, a modified GPE was adopted to construct a high energy density LMB, where Li-rich layered material was used as positive electrode with high working voltage up to 4.8 V and large specific capacity over 200 $mA\cdot h\cdot g^{-1}$ [26]. The quasi-solid state GPE was composed of a polymer substrate (polyvinylidene fluoride-co-hexafluoropropylene, PVDF-HFP) and a $LiPF_6$-based organic liquid electrolyte. It exhibited high self-diffusion ability, such that the GPE's ion conduction mechanism was similar to that of organic liquid electrolytes. To further ameliorate the ionic conductivity and mechanical properties of the GPE, a two-dimensional layered inorganic material, boron nitride (BN), was adopted as a functional nanofiller. BN is an electron insulator and an exceptional ion conductor and shows excellent thermodynamic and electrochemical stability that could improve the interfacial interactions with the polymer matrices. The LMB cell assembled with a

$Li_{1.18}C_{0.15}Ni_{0.15}Mn_{0.52}O_2$ (LLMO) positive and BN-(PVDF-HFP)-based GPE showed superior electrochemical performance and safety properties.

2. Experimental

2.1 Materials synthesis

A 1.0 g sample of PVDF-HFP particles was dissolved in 30 ml of acetone solution and then a certain amount of BN powder added and stirred for 12 h at room temperature. The BN amount in the PVDF-HFP was controlled at 0, 0.5, 1.0, 5.0 and 10.0 wt% and designated as pure PVDF-HFP, 0.5 wt% BN-(PVDF-HFP), 1.0 wt% BN-(PVDF-HFP), 5.0 wt% BN-(PVDF-HFP) and 10.0 wt% BN-(PVDF-HFP), respectively. The suspension was next transferred to a Teflon surface dish, dried at room temperature, and then dried at 60^{o}C for 24 h under vacuum to completely removed residual acetone and water. Finally, these thin films with different BN contents were cut into 18 mm diameter circles and soaked in 1 M $LiPF_6$ electrolyte for 24 h in solvent ethylene carbonate:dimethyl carbonate:ethyl methyl carbonate solvent (EC:DMC:EMC, respectively) with a molar ratio of 1:1:8.

The $Li_{1.18}Ni_{0.15}Co_{0.15}Mn_{0.52}O_2$ (LLMO) material was prepared by the sol-gel method, using Li_2CO_3, $Co(CH_3COO)_2$, $Ni(CH_3COO)_2$, and $Mn(CH_3COO)_2$ (analytical grade; Sinopharm Chemical Reagent Co., Ltd., Shanghai, China) as starting materials. The starting materials were mixed in a stoichiometric ratio of Li:Co:Ni:Mn of 1.18:0.15:0.15:0.52 and dissolved in deionized water with vigorous stirring. Then, citric acid was added at a Li:citric acid molar ratio of 1:2. The solution pH was adjusted to 7.3 by dropwise addition of aqueous ammonia and then stirred at 50 °C for 15 h. The resulting viscous solution was dried at 120 °C in a vacuum oven and the obtained substance was pretreated at 450 °C for 5 h before further use to eliminate the organic contents. Finally, the precursor was pressed into a pellet, annealed at 900 °C for 12 h, and quenched in liquid nitrogen.

2.2 Materials characterizations

The crystal structure of the material was studied by X-ray diffraction (XRD) on a Rigaku AXS D8 diffractometer with Cu K_α radiation. The morphology of the materials was studied by a JSM-6700F field emission scanning electron microscope (FESEM). Raman scattering was collected on a Thermo Scientific FT-Raman using Nd-line laser source. The mechanical properties of GPEs were measured using a universal testing machine (UTM, Shimadzu AG-I 1 kN). The amount of liquid electrolyte uptake was measured via their changes in weight before and after immersion in the 1 M $LiPF_6$ liquid electrolyte for 24 h. The electrolyte uptake value was calculated by the equation (1):

$$\text{Electrolyte uptake (\%)} = [(W_{after} - W_{before})/W_{before}] \times 100 \qquad (1)$$

where W_{before} and W_{after} are the weight of membrane before and after immersion in the liquid electrolyte, respectively. The thermal stability of the membranes with and without BN additive was studied by differential scanning calorimetry (DSC) on a TA-Q2000 (Waters LLC, USA) thermal analyzer using a heating rate of 10 ^oC min^{-1} over a temperature range of 50 - 250 ^oC.

2.3 Electrochemical experiments

Electrochemical experiments of the materials were performed on 2032-type coin cells using metallic Li as the negative electrode. The positive electrode was composed of 75 wt.% LLMO active material, 15 wt.% Super P conductive additive and 10 wt% polyvinylidene fluoride (PVDF) binder dissolved in N-methyl-2-pyrrolidone (NMP). The slurry mixture was coated on an Al current collector and dried overnight in a vacuum oven. The electrode was cut into 8 × 8 mm^2 for use and the electrode thickness and density were about 30 μm and 1.3 g cm^{-3}, respectively. It is noted that the above values were identical for all samples. The obtained LLMO material, electrolyte (x) and Li metal were assembled together to construct LLMO/electrolyte (x)/Li cells, where x = 1 mol·L^{-1} $LiPF_6$, pure PVDF-HFP, 0.5 wt% BN-(PVDF-HFP) and 10.0 wt% BN-(PVDF-HFP), respectively. Galvanostatic charge-discharge experiments were performed on a Land-2100 automatic battery tester. The time evolution of voltage profiles was measured on a symmetric Li/x/Li cell at a current density of 1.0 mA cm^{-2} and the polarity was reversed for every 1 h. Cyclic voltammetry (CV) was performed on a Bio-Logic VSP multichannel potentiostatic-galvanostatic system with a scan rate of 0.1 mV s^{-1}. The ionic conductivities of GPEs at the range of 25-80 ^{o}C were determined on the Bio-Logic VSP instrument in the frequency range of 0.1 Hz to 1.0 MHz with a 10 mV AC amplitude. The samples for the measurements were prepared by sandwiching the GPEs between two stainless-steel electrodes in a 2032 coin cell. The ionic conductivity (σ) of GPEs was calculated from the following equation (2):

$$\sigma = (1/R) \times (d/A) \tag{2}$$

where R is the electrolyte resistance, d is the thickness of electrolyte, and A is the area of the electrode [7].

3. Results and discussion

The schematic diagram for the preparation of pure PVDF-HFP and 0.5 and 10.0 wt% BN-(PVDF-HFP) membranes is shown in **Fig. 1**. These membranes were cut into 18 mm diameter circles (**Fig. S1**, Supporting Information) and, because of the presence of white BN powders, the transparency of the latter two membranes were decreased. SEM images of the three membranes showed that the BN filler was well dispersed in the polymer matrices (**Fig. S2**) and the luminosity a bit lower, which was indicative of BN's insulation in the polymer electrolyte membranes. In addition, when the BN content increased from 0.5 to 10.0 wt%, film graininess clearly increased. Cross-sectional SEM images clearly showed that the three film thicknesses were ~40 μm and with similar morphology (**Fig. S3**).

The crystal structures of pure PVDF-HFP and 0.5 and 10.0 wt% BN-(PVDF-HFP) polymer substrates were characterized by XRD (**Fig. 2a**). For pure PVDF-HFP, XRD displayed five broad major reflections at 17.8, 19.8, 21.7, 26.8, and 38.9^{o} (2θ), which corresponded to the 100, 020, 110, 021, and 002 crystal planes of PVDF, respectively. In addition, a broader reflection was observed at 18.6^{o}, which corresponded to the amorphous phase of hexafluoropropene [27, 28]. Therefore, the prepared PVDF-HFP polymer was regarded as a semicrystalline phase. The intensities of XRD reflections decreased as BN content increased up to 0.5 and 10.0 wt%, because the BN additive significantly disrupted the orderly alignment of PVDF-HFP chains, resulting in the amorphous phase. Generally, we believed that the emergence of amorphous phase was a sign of improved ionic conductivity [21, 23], which was more conducive to Li^+ transmission. To further confirm the reduction in crystallinity after adding different amounts of BN, we also conducted DSC analysis as shown in **Fig. S4**. The DSC showed that the melting temperature of the pure PVDF-HFP membrane was around 126 ^{o}C, while the melting temperature of BN-modified membranes were slightly reduced. Additionally, the degree of crystallinity of each polymer membrane was calculated with following formula (3):

$$\chi_c = \Delta H_m / \Delta H_{m*} \tag{3}$$

Where χ_c was degree of crystallinity, ΔH_m represented the enthalpy of fusion of membrane sample and ΔH_{m*} represented the enthalpy of PVDF-HFP polymer (i.e.104.7 J g^{-1}) [27]. It could be inferred that the PVDF-HFP with BN fillers showed higher thermal stability, lower melting enthalpy, and less crystallinity, indicating the increase of amorphous phase. Raman scattering was used to verify the existence of BN additive and the peak at 1367 cm^{-1} Raman spectra demonstrated the BN additive in both 0.5 and 10.0 wt% BN-(PVDF-HFP), which corresponded to the E_{2g} phonon mode of B-N bonds (**Fig. 2b**). Therefore, BN was successfully introduced into the polymer substrate.

Gelatinized polymer electrolytes were prepared by immersing the three membranes, 0, 0.5 and 10.0 wt% BN-(PVDF-HFP) in 1 M $LiPF_6$-based electrolyte for 24 h. A test on absorptivity was performed to verify electrolyte infiltration. Images of three samples before and after soaking in the liquid electrolyte revealed that the soaked membranes in the diameter direction expanded to different degrees (18.0 to 18.5, 18.0 to 18.9, and 18.0 to 18.7 mm/mm for 0, 0.5, and 10.0 wt% BN-(PVDF-HFP), respectively (**Fig. S5**). The electrolyte uptakes of 0, 0.5, and 10.0 wt% BN-(PVDF-HFP) were 39, 120, and 89 %, respectively. It was concluded that 0.5 wt% BN-(PVDF-HFP) manifested the best infiltration effects. This was ascribed to aligned polymer chains being randomly interrupted, thereby providing more active sites. When the BN content was 10 wt%, the electrolyte uptake decreased to 89%, thus impairing ionic conductivity, because of inhomogeneous dispersion and aggregation of excessive BN.

The influence of temperature on of GPEs' ionic conductivities was also studied for pure and BN-modified membrane. It showed that the ionic conductivities of the two BN-containing membranes increased with increased temperature from 25 to 80 ^{o}C at the same time interval (**Fig. 3**). This illustrated that ionic conductivity favorably increased along with the temperature increase due to faster diffusion of dissociated Li^+ at higher temperatures. The 0.5 wt% BN-(PVDF-HFP) membrane exhibited lower bulk resistance and higher ionic conductivity (**Fig. 3a** and **3b**, respectively) in comparison with the pure PVDF-HFP membrane, which was due to that BN addition greatly contributed to faster electron and ion migration. On one hand, 0.5 wt% BN nanofillers were evenly distributed into PVDF-HFP instead of obvious aggregation (such as SiO_2 and Al_2O_3 [29], requiring superfluous amount to effectively inhibit the growth of Li dendrites, whereas sacrificing the ionic conductivity), which profitably increased the movement of the lithium ions. On the other hand, BN with Lewis acidic traits could interact with other Lewis bases and thereby increasing the ionic conductivity of GPEs by trapping the anions of the electrolytes. These features drove BN as a promising candidate for GPE additives.

Mechanical strength is another vital index for the practical application of GPEs, particularly in large scale energy storage systems or electric vehicles. The mechanical strength of pure PVDF-HFP and 0.5 wt% BN-(PVDF-HFP) was characterized by Young's modulus and tensile strength (**Fig. 4**). The results here showed that the Young's modulus (560 MPa) and tensile strength (35 MPa) of 0.5 wt% BN-(PVDF-HFP) were 1.7 and 1.5-times larger than those of pure PVDF-HFP, respectively. The interface between two-dimensional layered material BN and the polymer substrate can couple certain interactions [29-33], and a small amount of BN can significantly improve the mechanical properties of polymer matrix. Furthermore, from rheological tests, the energy storage moduli of the two membranes were found to be larger than the loss moduli, confirming that both samples were rigid materials (**Fig. 4b**). Comparing the ratio of the energy storage and loss moduli, the ratio of pure PVDF-HFP was significantly larger than that of 0.5 wt% BN-(PVDF-HFP), indicating that the polymer matrix with BN was able to resist more deformation under violent forces, thereby favorably inhibiting lithium dendrite puncture to improve battery safety.

The growth of lithium dendrites in symmetric Li/x/Li cells (x = 1 M $LiPF_6$, 0, 0.5, and 10.0 wt% BN-(PVDF-HFP)) were compared by performing galvanostatic lithium plating/stripping at the current density of 1 mA·cm^{-2} (**Fig. 5**). The Li/1 M $LiPF_6$/Li cell showed short-circuit because of continuous lithium dendrite formation, which was affirmed by the sharp rise in voltage hysteresis with cycle time (180 h) and originated from the lower mechanical strength of the commercial Celgard 2350 separator [29]. In contrast, the Li/x/Li cell with GPEs containing BN exhibited a less overpotential and longer cycling ability (e.g., 900 h for 0.5 wt% BN-(PVDF-HFP)), which was attributed to high ionic conductivity and large mechanical modulus of the GPEs.

Subsequently, electrochemical measurements were performed using LLMO/x/Li cells (x = 1 M $LiPF_6$, 0, 0.5 and 10.0 wt% BN-(PVDF-HFP)) at a 0.2 C rate (We also tested the electrochemical performance of cells using 1.0 and 5.0 wt% BN-(PVDF-HFP) membranes as shown in **Fig. S6**. Therefore, the samples with the maximum contrast were chose to intuitively illustrate the related problems, i.e. x = 1 M $LiPF_6$, 0, 0.5 and 10.0 wt% BN-(PVDF-HFP)). All the cells displayed very similar initial charge and discharge profiles (**Fig. 6a**), which suggested that BN additive in GPEs did not change the reaction mechanisms of the LLMO material. The LLMO/1 M $LiPF_6$/Li cell delivered an initial discharge capacity of 253 mA·h·g^{-1}. For LLMO/pure PVDF-HFP/Li, the initial discharge capacity was only 231 mA·h·g^{-1}, which was probably caused by decreased ion conductivity of the quasi-solid state electrolyte in pure PVDF-HFP. The LLMO/0.5 wt% BN-(PVDF-HFP)/Li cell showed an initial discharge capacity of 263 mA·h·g^{-1}, which was slightly higher than that in 1

M $LiPF_6$. However, the LLMO/10 wt% BN-(PVDF-HFP)/Li cell only delivered an initial capacity of 226 mA·h·g^{-1}. The cell assembled with 0.5 wt% BN-(PVDF-HFP) membrane had the highest initial discharge capacity, which was due to BN's high ionic conductivity, thereby promoting Li^+ transportation and better performance. However, when the BN content was 10 wt%, the discharge capacity was significantly reduced because the excessive amount of BN additive easily caused particle agglomeration and hindered Li^+ transportation. The cycling stability of the cells at a 0.5 C rate showed that the specific capacities of LLMO/1 M $LiPF_6$/Li rapidly decreased and the capacity retention was only 30% after 300 cycles (**Fig. 6b**). In comparison, the cycling stability of LLMO/pure PVDF-HFP/Li, LLMO/0.5 wt% BN-(PVDF-HFP)/Li, and LLMO/10 wt% BN-(PVDF-HFP)/Li cells were significantly improved and the capacity retention after 300 cycles at 71, 79, and 64%, respectively. The use of GPEs reduced the side effects of liquid electrolytes at high voltage, thus improving the batteries' cycling stability. The cycling performance of the cells at 55 ^{o}C with a 0.5 C rate showed that the attenuation of LLMO/1 M $LiPF_6$/Li cell was quite serious and the capacity retention was only 74% after 50 cycles, while the cell assembled with 0.5 wt% BN-(PVDF-HFP) showed excellent cycling performance, with capacity retention up to 98% (**Fig. 6d**). This was attributed to the higher ionic conductivity and mechanical strength of the BN additive, playing a significant regulatory role in lithium deposition and forming a stable SEI film on the lithium metal side. In addition, we also conducted the electrochemical cycling at low temperature (i.e. 0 ^{o}C) as shown in **Fig. S7**, it could be observed that LLMO/1 M $LiPF_6$/Li cell exerted severe capacity degradation at 0.5 C after 50 cycles, while the cell assembled with 0.5 wt% BN-(PVDF-HFP) revealed impressive cycling performance apart from the slight reduction of the first several cycles. Therefore, BN-modified GPEs could be potentially applied in a wide temperature range. In order to confirm that the growth of Li dendrites was indeed suppressed, the SEM images of Li metal surface was performed after cycling as displayed in **Fig. S8**. The Li metal surface in the cell with 0.5 wt% BN-(PVDF-HFP) maintained slight cracks but smooth and flat morphology (**Fig. S8b**), whereas that in the cell with 1 M $LiPF_6$ (Celgard-2350) occurred noticeable roughness and slender dendrites (**Fig. S8a**). This could be attributed to the synergistic combination of increased mechanical modulus and high ionic conductivity of BN additive. The effects of GPEs with and without BN on the rate performance of LLMO were examined by performing the charge-discharge experiments at different cycle rates from 0.2 C (40 $mA \cdot g^{-1}$) to 10 C (2 $A \cdot g^{-1}$) as shown in **Fig. 6c**. The LLMO/0.5 wt% BN-(PVDF-HFP)/Li cell displayed the best rate performance. At 5 and 10 C current rates, the discharge capacities of the LLMO/0.5 wt% BN-(PVDF-HFP)/Li cell were 137 and 105 $mA \cdot h \cdot g^{-1}$, while the corresponding capacities of LLMO/1 M $LiPF_6$/Li were only 117 and 67 $mA \cdot h \cdot g^{-1}$, respectively. Notably, the cell assembled with 10 wt% BN-(PVDF-HFP) showed inferior performance compared with LLMO/1 M $LiPF_6$/Li, which might have been due to agglomerated BN particles affecting Li^+ transportation. Therefore, minimal addition of BN synergistically facilitated Li^+ transfer and repressed dendrites, thereby significantly enhancing the electrochemical performance and safety of the LMBs.

CV measurements at a scan rate of 0.1 $mV \cdot s^{-1}$ exhibited two current peaks in the oxidation process (**Fig. S9**) for the first cycle of the LLMO/1 M $LiPF_6$/Li cell. The one at 4.07 V was attributed to Li^+ extraction from the lithium layer of LLMO with the oxidation of Ni^{2+} to Ni^{4+} and Co^{3+} to $Co^{3.6+}$ [34, 35]. The other one at ~4.58 V was due to the simultaneous Li^+ extraction from the lithium layer and transition metal layer, combined with concurrent oxygen release and layer-to-spinel structure transformation. In the reduction process, the peaks at 4.41, 3.71, and 3.25 V were attributed to the reduction of $Co^{3.6+}$, Ni^{4+}, and Mn^{4+}, respectively. The subsequent CV curves showed clear changes with respect to the first cycle. Absence of the oxidic peak at ~4.58 V indicated that the oxygen loss process was completely finished in the first cycle. In addition, the CV curves of the other three cells were analogous to that of the LLMO/1 M $LiPF_6$/Li cell, which convincingly verified that the reaction mechanism of LLMO was not affected by the application of GPEs with and without BN additive. Moreover, the overlapping of CV curves of the LLMO/0.5 wt%BN-(PVDF-HFP)/Li cell indicated dramatic improvements in the cell stability compared with the LLMO/1 M $LiPF_6$/Li cell, which was consistent with the electrochemical analysis (**Fig. 6**).

The morphology of the LLMO electrodes was also examined by disassembling the cells after 200 cycles. The LLMO material assembled with 1 M $LiPF_6$ had serious adhesion between the active electrode and Al current collector (**Fig. 7**). The electrode surface was extremely rough with visible cracks and this harsh surface/interface might have resulted in the formation of uneven SEI films, which might then have blocked the ion diffusion and electron transportation. In contrast, the LLMO electrode in the LLMO/0.5 wt% BN-(PVDF-HFP)/Li cell retained a smooth and intact morphology, which was more conducive for improving the holistic properties of LMBs.

4. Conclusions

In conclusion, two-dimensional BN that donated multipurpose characteristics to a PVDF-HFP-based GPE for LMBs was successfully distributed into the GPEs and used in lithium metal batteries together with Li-rich layered oxide postivie electrode. The GPE with 0.5 wt% BN additive exhibited much high ionic conductivity and mechanical strength. Thereby, BN additive significantly promoted ion diffusion and, more importantly, was favorable for resisting lithium dendrite punctures by virtue of a mechanically robust blocking layer. The above advantages gave the present LLMO/0.5 wt% BN-GPEs/Li cell superior stability, rate capability, and high safety, in particular, at high temperature 55 ^{o}C.

Appendix A. Supplementary data

Supplementary data to this article can be found online at ……….

Conflict of interest statement

The authors declare no competing financial interest.

Acknowledgements

This work was financially supported by the Start-up Grants from Dongguan University of Technology for the High-level Talents (KCYKYQD2017016) and Research Start-up Funds of DGUT (No. GC300501-118 and GC300501-114).

References

[1] X. Wang, H. Zhai, B. Qie, Q. Cheng, A. Li, J. Borovilas, B. Xu, C. Shi, T. Jin, X. Liao, Y. Li, X. He, S. Du, Y. Fu, M. Dontigny, K. Zaghib, Y. Yang, Rechargeable solid-state lithium metal batteries with vertically aligned ceramic nanoparticle/polymer composite electrolyte, Nano Energy 60 (2019) 205-212.

[2] D. Lin, Y. Liu, Y. Cui, Reviving the lithium metal anode for high-energy batteries, Nat. Nanotechnol. 12 (2017) 194-206.

[3] H.-G. Wang, S. Yuan, D.-L. Ma, X.-B. Zhang, J.-M. Yan, Electrospun materials for lithium and sodium rechargeable batteries: from structure evolution to electrochemical performance, Energy Environ. Sci. 8 (2015) 1660-1681.

[4] S. Li, X. Meng, Q. Yi, J.A. Alonso, M.T. Fernández-Díaz, C. Sun, Z.L. Wang, Structural and electrochemical properties of $LiMn_{0.6}Fe_{0.4}PO_4$ as a cathode material for

flexible lithium-ion batteries and self-charging power pack, Nano Energy 52 (2018) 510-516.

[5] S. Li, D. Zhang, X. Meng, Q.-A. Huang, C. Sun, Z.L. Wang, A flexible lithium-ion battery with quasi-solid gel electrolyte for storing pulsed energy generated by triboelectric nanogenerator, Energy Storage Materials 12 (2018) 17-22.

[6] M. Zhou, Y. Lyu, Y. Liu, B. Guo, Porous scaffold of TiO_2 for dendrite-free lithium metal anode, J. Alloys Compd. 791 (2019) 364-370.

[7] Q. Yi, W. Zhang, S. Li, X. Li, C. Sun, Durable sodium battery with a flexible $Na_3Zr_2Si_2PO_{12}$-PVDF-HFP composite electrolyte and sodium/carbon cloth anode, ACS Appl. Mater. Interfaces 10 (2018) 35039-35046.

[8] K. Zhang, G.-H. Lee, M. Park, W. Li, Y.-M. Kang, Recent developments of the lithium metal anode for rechargeable non-aqueous batteries, Adv. Energy Mater. 6 (2016) 1600811.

[9] W. Zhang, J. Nie, F. Li, Z.L. Wang, C. Sun, A durable and safe solid-state lithium battery with a hybrid electrolyte membrane, Nano Energy 45 (2018) 413-419.

[10] S. Choudhury, L.A. Archer, Lithium fluoride additives for stable cycling of lithium batteries at high current densities, Adv. Electron. Mater. 2 (2016) 1500246.

[11] F. Ding, W. Xu, G.L. Graff, J. Zhang, M.L. Sushko, X. Chen, Y. Shao, M.H. Engelhard, Z. Nie, J. Xiao, X. Liu, P.V. Sushko, J. Liu, J.G. Zhang, Dendrite-free lithium deposition via self-healing electrostatic shield mechanism, J. Am. Chem. Soc. 135 (2013) 4450-4456.

[12] E. Kazyak, K.N. Wood, N.P. Dasgupta, Improved cycle life and stability of lLithium metal anodes through ultrathin atomic layer deposition surface treatments, Chem. Mater. 27 (2015) 6457-6462.

[13] D. Lin, Y. Liu, Z. Liang, H.W. Lee, J. Sun, H. Wang, K. Yan, J. Xie, Y. Cui, Layered reduced graphene oxide with nanoscale interlayer gaps as a stable host for lithium metal anodes, Nat. Nanotechnol. 11 (2016) 626-632.

[14] D. Lin, D. Zhuo, Y. Liu, Y. Cui, All-integrated bifunctional separator for Li dendrite detection via novel solution synthesis of a thermostable polyimide separator, J. Am. Chem. Soc. 138 (2016) 11044-11050.

[15] Q.C. Liu, T. Liu, D.P. Liu, Z.J. Li, X.B. Zhang, Y. Zhang, A flexible and wearable lithium-oxygen battery with record energy density achieved by the interlaced architecture inspired by bamboo slips, Adv. Mater. 28 (2016) 8413-8418.

[16] T. Liu, Q.C. Liu, J.J. Xu, X.B. Zhang, Cable-type water-survivable flexible Li-O_2 battery, Small 12 (2016) 3101-3105.

[17] Y. Lu, Z. Tu, L.A. Archer, Stable lithium electrodeposition in liquid and nanoporous solid electrolytes, Nat. Mater. 13 (2014) 961-969.

[18] G. Zheng, S.W. Lee, Z. Liang, H.W. Lee, K. Yan, H. Yao, H. Wang, W. Li, S. Chu, Y. Cui, Interconnected hollow carbon nanospheres for stable lithium metal anodes, Nat. Nanotechnol. 9 (2014) 618-623.

[19] A. Chandra, Hot-pressed PEO-PVP blended solid polymer electrolytes: ion transport and battery application, Polym. Bull. 73 (2016) 2707-2718.

[20] S. Das, A. Ghosh, Ionic relaxation in PEO/PVDF-HFP-$LiClO_4$ blend polymer electrolytes: dependence on salt concentration, J. Phys. D. Appl. Phys. 49 (2016) 235601.

[21] X. Zhang, S. Zhao, W. Fan, J. Wang, C. Li, Long cycling, thermal stable, dendrites free gel polymer electrolyte for flexible lithium metal batteries, Electrochim. Acta 301 (2019) 304-311.

[22] J.F. Vélez, M. Aparicio, J. Mosa, Covalent silica-PEO-LiTFSI hybrid solid electrolytes via sol-gel for Li-ion battery applications, Electrochim. Acta 213 (2016) 831-841.

[23] J. Yang, Z. Huang, B. Huang, J. Zhou, X. Xu, Influence of phosphorus sources on lithium ion conducting performance in the system of $Li_2O–Al_2O_3–GeO_2–P_2O_5$ glass–ceramics, Solid State Ionics 270 (2015) 61-65.

[24] W. Zhou, H. Gao, J.B. Goodenough, Low-cost hollow mesoporous polymer spheres and all-solid-state lithium, sodium batteries, Adv. Energy Mater. 6 (2016) 1501802.

[25] Q. Xiao, C. Deng, Q. Wang, Q. Zhang, Y. Yue, S. Ren, In situ cross-linked gel polymer electrolyte membranes with excellent thermal stability for lithium ion batteries, ACS Omega 4 (2019) 95-103.

[26] M. Xu, L. Fei, W. Lu, Z. Chen, T. Li, Y. Liu, G. Gao, Y. Lai, Z. Zhang, P. Wang, H. Huang, Engineering hetero-epitaxial nanostructures with aligned Li-ion channels in Li-rich layered oxides for high-performance cathode application, Nano Energy 35 (2017) 271-280.

[27] I. Kim, B.S. Kim, S. Nam, H.J. Lee, H.K. Chung, S.M. Cho, T.H.T. Luu, S. Hyun, C. Kang, Cross-linked poly(vinylidene fluoride-co-hexafluoropropene) (PVDF-co-HFP) gel polymer electrolyte for flexible Li-ion battery integrated with organic light emitting diode (OLED), Mater. 11 (2018) 11040543.

[28] P. Senthil Kumar, A. Sakunthala, M.V. Reddy, M. Prabu, Structural, morphological, electrical and electrochemical study on plasticized PVdF-HFP/PEMA blended polymer electrolyte for lithium polymer battery application, Solid State Ionics 319 (2018) 256-265.

[29] J. Shim, H.J. Kim, B.G. Kim, Y.S. Kim, D.-G. Kim, J.-C. Lee, 2D boron nitride nanoflakes as a multifunctional additive in gel polymer electrolytes for safe, long cycle life and high rate lithium metal batteries, Energy Environ. Sci. 10 (2017) 1911-1916.

[30] K.H. Oh, D. Lee, M.J. Choo, K.H. Park, S. Jeon, S.H. Hong, J.K. Park, J.W. Choi, Enhanced durability of polymer electrolyte membrane fuel cells by functionalized 2D boron nitride nanoflakes, ACS Appl. Mater. Interfaces 6 (2014) 7751-7758.

[31] A. Pakdel, Y. Bando, D. Golberg, Nano boron nitride flatland, Chem. Soc. Rev. 43 (2014) 934-959.

[32] S.Y. Xie, W. Wang, K.A. Fernando, X. Wang, Y. Lin, Y.P. Sun, Solubilization of boron nitride nanotubes, Chem. Commun. (2005) 3670-3672.

[33]Y. Lin, T.V. Williams, J. Connell, Soluble, exfoliated hexagonal boron nitride nanosheets, J. Phys. Chem. Lett. 1 (2010) 277-283.

[34] X. Bian, Q. Fu, X. Bie, P. Yang, H. Qiu, Q. Pang, G. Chen, F. Du, Y. Wei, Improved electrochemical performance and thermal stability of Li-excess $Li_{1.18}Co_{0.15}Ni_{0.15}Mn_{0.52}O_2$ cathode material by Li_3PO_4 surface coating, Electrochim. Acta 174 (2015) 875-884.

[35] X. Bian, Q. Fu, Q. Pang, Y. Gao, Y. Wei, B. Zou, F. Du, G. Chen, Multi-functional surface engineering for Li-excess layered cathode material targeting excellent electrochemical and thermal safety properties, ACS Appl. Mater. Interfaces 8 (2016) 3308-3318.

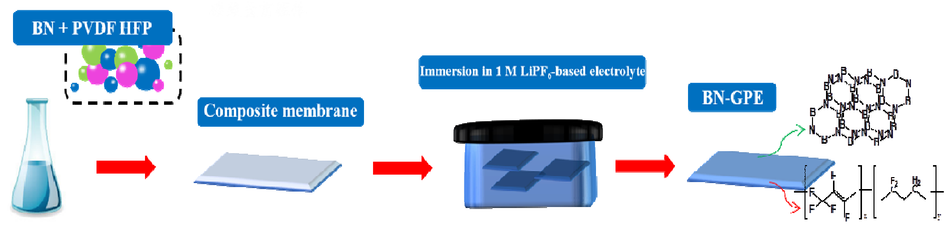


Fig. 1. Schematic of the preparation of pure PVDF-HFP and 0.5 and 10.0 wt% BN-(PVDF-HFP) membranes.

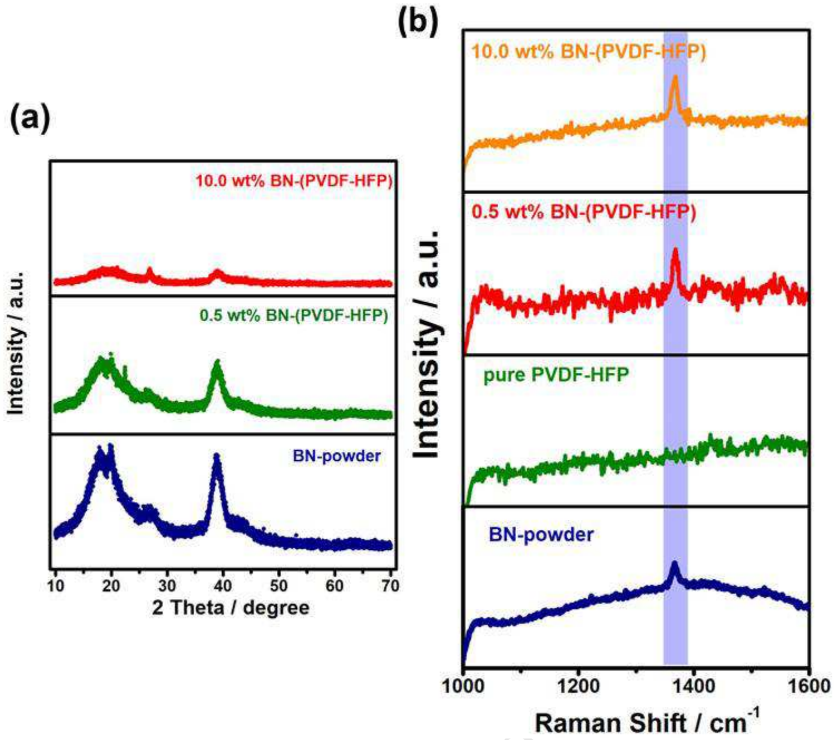


Fig. 2. XRD patterns of pure PVDF-HFP and 0.5 and 10.0 wt% BN-(PVDF-HFP) membranes (a) and Raman patterns of BN-powder, pure PVDF-HFP, and 0.5 and 10.0 wt% BN-(PVDF-HFP) membranes (b)

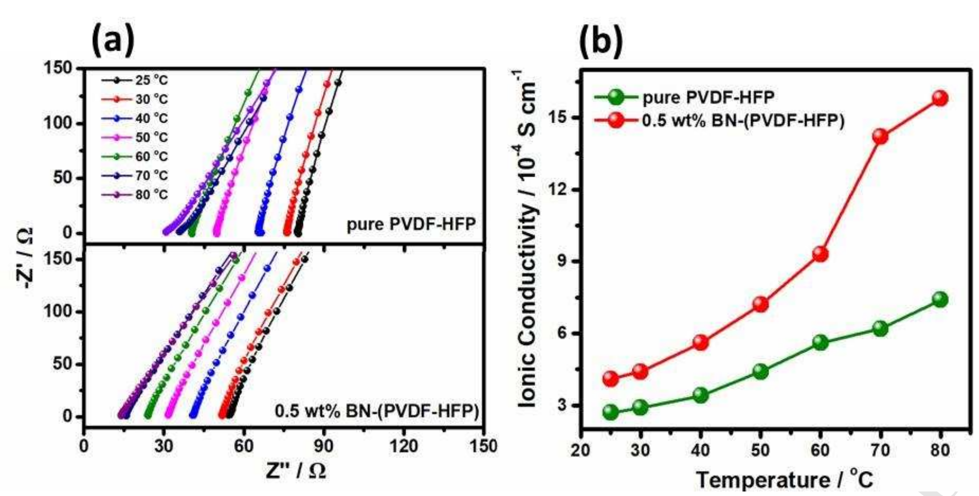


Fig. 3. Nyquist plots (a) and ionic conductivities (b) of pure PVDF-HFP and 0.5 wt% BN-(PVDF-HFP) membranes at different temperatures.

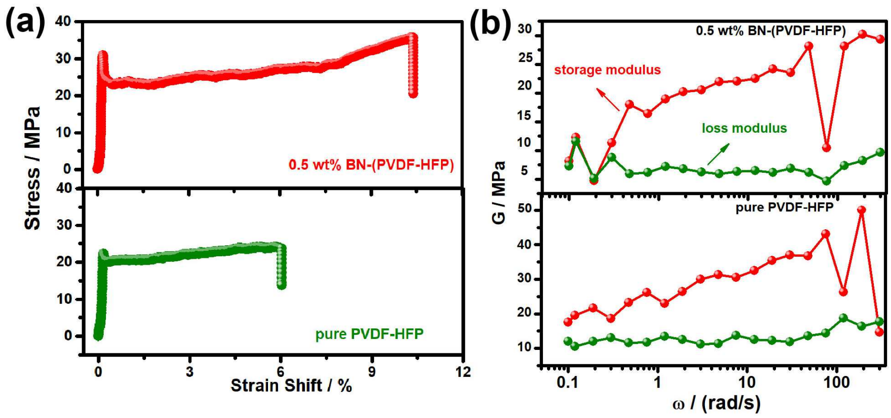


Fig. 4. Tensile curves (a) and flowing deformation curves (b) of pure PVDF-HFP and 0.5 wt% BN-(PVDF-HFP) membranes.

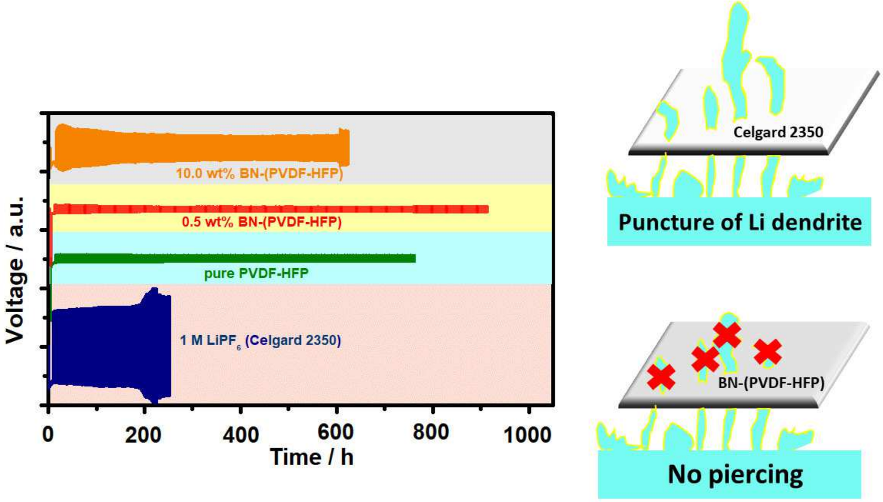


Fig. 5. Li metal deposition profiles of symmetric Li/x/Li cells with x=1 M $LiPF_6$, pure PVDF-HFP and 0.5 and 10.0 wt% BN-(PVDF-HFP).

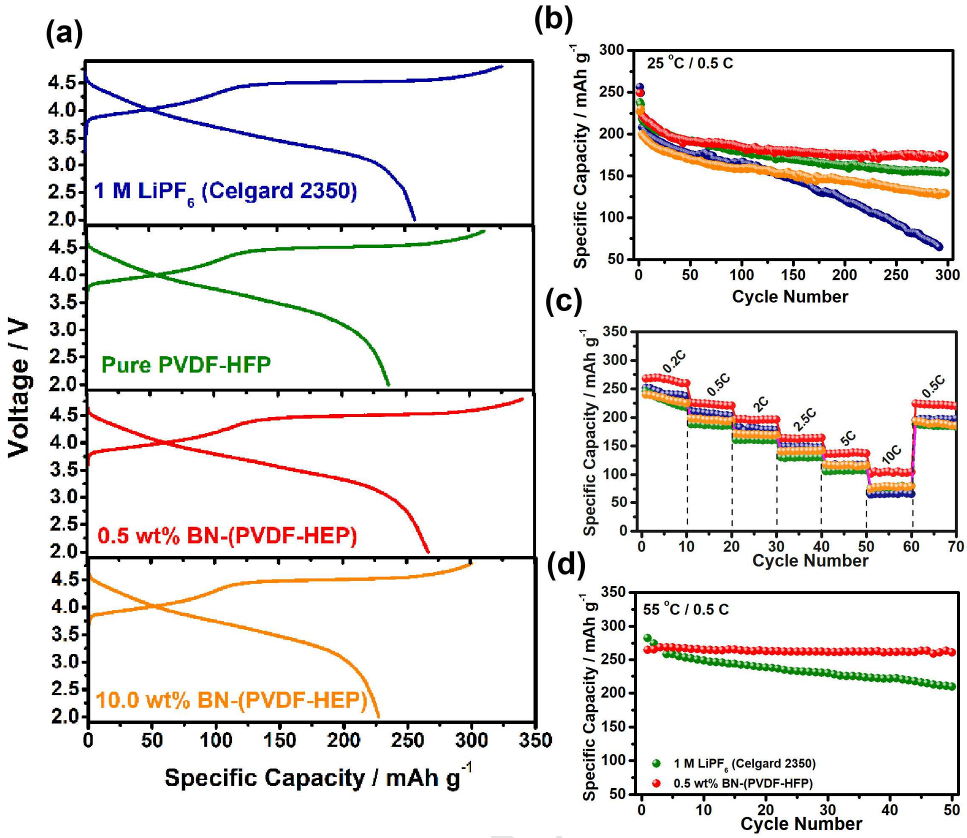


Fig. 6. First charge-discharge curves at 0.2 C (a), cycling performance at 0.5 C (b), rate performance at 25 °C (c), and cycling performance at 0.5 C at 55 °C (d) of LLMO/x/Li cells with x=1 M $LiPF_6$, pure PVDF-HFP and 0.5 and 10.0 wt% BN-(PVDF-HFP).

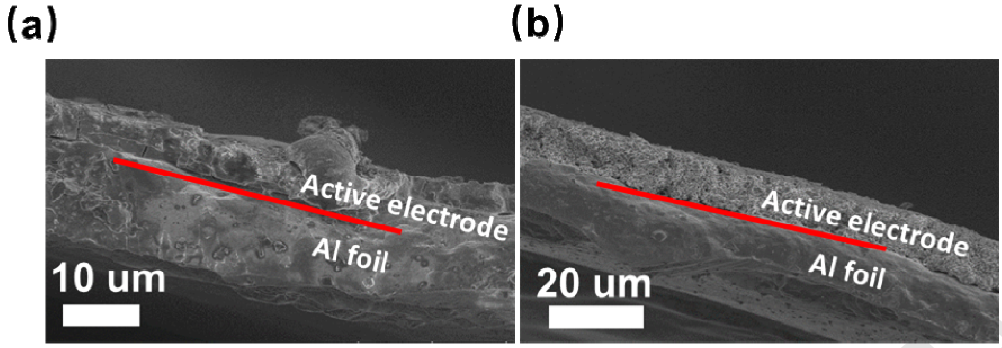


Fig. 7. SEM images of active electrodes using 1 M $LiPF_6$ (a) and 0.5 wt% BN-(PVDF-HFP) (b) electroltyes after 200 cycles.

1. We have successfully incorporated 2D BN fillers into a gel polymer electrolyte based on PVDF-HFP.
2. Only 0.5 wt% addition of BN can readily obtain a comparative ionic conductivity and improved mechanical modulus.
3. The $Li_{1.18}Ni_{0.15}Co_{0.15}Mn_{0.52}O_2$/0.5 wt% BN-(PVDF-HFP)/Li cell performs a superior stability and rate capability.